

Title: Deepfake Detection Using Convolutional Neural Networks: A Baseline Study

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Abstract:

This paper presents a binary classification system for detecting deepfake videos using a pre-trained MobileNetV2 convolutional neural network. We utilized a subset of the FaceForensics++ dataset, extracting 2000 frames from 400 videos. Our baseline model achieved 71.25% test accuracy. Robustness testing revealed that the model is more sensitive to Gaussian noise than JPEG compression.

1. Introduction

The rapid advancement of deep learning has enabled the creation of highly realistic manipulated videos, commonly known as deepfakes. These synthetic media pose significant threats to information integrity, personal privacy, and public trust. Detecting such manipulations has become a critical research challenge.

In this work, we propose a lightweight CNN-based approach using MobileNetV2 for binary classification of real and fake facial images. Our system is trained on a subset of the FaceForensics++ benchmark dataset and evaluated under various degradation conditions to assess robustness.

2. Related Work

Several approaches have been proposed for deepfake detection. Rossler et al. introduced FaceForensics++, a large-scale dataset for facial manipulation detection. MesoNet, proposed by Afchar et al., uses a shallow CNN architecture focusing on mesoscopic properties of images. XceptionNet, adapted from image classification, has shown strong performance on deepfake detection tasks. Our work builds on these findings by applying transfer learning with MobileNetV2, a lightweight architecture suitable for resource-constrained environments.

3. Methodology

3.1 Dataset

We utilized a subset of the FaceForensics++ dataset consisting of 400 videos — 200 real and 200 fake. Five frames were extracted from each video, resulting in 2000 images total (1000 real, 1000 fake). All frames were resized to 224×224 pixels.

3.2 Preprocessing

Images were normalized using ImageNet mean and standard deviation values (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]). The dataset was split into 80% training (1600 images) and 20% testing (400 images).

3.3 Model Architecture

We employed MobileNetV2 pre-trained on ImageNet, replacing the final classification layer with a binary output layer (real/fake). The model was trained using the Adam optimizer with a learning rate of 0.001 and Cross-Entropy loss for 5 epochs.

4. Experiments and Results

4.1 Baseline Performance

The model was trained for 5 epochs on CPU. Training accuracy improved consistently across epochs, reaching 79.75% by the final epoch. Test accuracy was 71.25%, indicating reasonable generalization for a baseline model.

Epoch	Loss	Train Accuracy
1	0.6302	63.44%
2	0.5163	73.44%
3	0.4955	76.19%
4	0.4619	77.56%
5	0.4191	79.75%

4.2 Robustness Testing

To evaluate model robustness, we tested the model under two types of degradation: JPEG compression and Gaussian noise.

Condition	Accuracy
No degradation	71.25%
JPEG Quality 50	70.50%
JPEG Quality 20	65.75%
Noise level 10	56.00%
Noise level 25	58.25%

Results show that the model maintains reasonable accuracy under JPEG compression but degrades significantly under Gaussian noise, dropping to 56.00% at noise level 10.

5. Discussion

The baseline model demonstrates promising results for deepfake detection using a lightweight architecture. The sensitivity to Gaussian noise suggests that further data augmentation during training could improve robustness. JPEG compression has minimal impact, which is encouraging given that most real-world images undergo some form of compression.

6. Conclusion and Future Work

In this paper, we presented a baseline deepfake detection system using MobileNetV2 transfer learning on the FaceForensics++ dataset. Our model achieved 71.25% test accuracy with minimal preprocessing and training resources.

Future work will focus on:

- Hyperparameter tuning to improve accuracy
- Data augmentation to improve robustness against noise
- Error analysis to identify misclassified samples
- Expanding the dataset size for better generalization

References

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